

FINAL EXPLANATION: MATHEMATICS GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: MATHEMATICS GRADE 10 | SUB-STRAND 1.3: QUADRATIC EQUATIONS | FINAL EXPLANATION

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

Final Explanation: Planning the Githurai Harambee Hall. Driving question: How can we use quadratic equations to find unknown dimensions and quantities that help our community solve real planning problems, like building the Harambee hall in Githurai? Use what you observed and learned in Lessons 1 to 7 to answer each question in the space under it. Show every step of your working, write in full sentences when you explain, and always say what your answer means for the community. Use the Githurai hall data: the floor is a rectangle with an area of 120 m^2 , the width is w and the length is $w + 2$. You may use a scientific calculator for Section 4.

Section 1: Why a linear equation is not enough

The Githurai committee wants a rectangular hall floor of area 120 m^2 , with the length 2 m more than the width w . (a) Write an equation for the area in terms of w . (b) Expand it and write it in the form $ax^2 + bx + c = 0$. State the values of a , b and c . (c) Explain why trial-and-error and linear equation methods did not work cleanly for the committee. How is this equation different from a linear equation?

(a) Area = length $\times$ width, so $w(w + 2) = 120$. (b) Expanding with the distributive law: $w^2 + 2w = 120$. Moving 120 to the left: $w^2 + 2w - 120 = 0$. This is in the form $ax^2 + bx + c = 0$ with $a = 1$, $b = 2$ and $c = -120$. (c) Trial-and-error only gave us a table of (w , Area) pairs and we had to keep guessing. A linear equation cannot be used because w is multiplied by itself, which produces a w^2 term. We cannot isolate w with one-step inverse operations such as adding, subtracting, multiplying or dividing. A linear equation has w only to the power 1, while a quadratic equation has a highest power of 2. A quadratic equation is therefore needed, with new methods of solving it.

Section 2: Solving the hall problem by factorisation

(a) Factorise $w^2 + 2w - 120 = 0$. Explain how you chose your two numbers. (b) Use the Zero Product Property to find the values of w . (c) Which value will the committee use for the width, and why? State the length of the hall. (d) Check that

(a) I need two numbers that multiply to -120 and add to $+2$. These are $+12$ and -10 , because $12 \times (-10) = -120$ and $12 + (-10) = 2$. So $(w + 12)(w - 10) = 0$. (b) Zero Product Property: if two factors multiply to give zero, at least one factor must equal zero. So $w + 12 = 0$ gives $w = -12$, or $w - 10 = 0$ gives $w = 10$. (c) The width is a physical measurement, so a negative value is rejected. The committee should use $w = 10 \text{ m}$. The length is $w + 2 = 12 \text{ m}$. (d) Check: $10 \times 12 = 120 \text{ m}^2$, which matches the required area. Both $w = 10$ and $w = -12$ make the equation true, but only $w = 10$ makes sense for the hall.

your dimensions give an area of 120 m ² .	
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Section 3: Three methods, one hall

(a) Show how you would solve $w^2 + 2w - 120 = 0$ by completing the square. (b) Show how you would solve it using the quadratic formula. (c) Compare the three methods you have learned (factorisation, completing the square and the quadratic formula). When is each most useful? Use the Nakuru school garden equation $x^2 + 6x = 48$ or another example from class to support your answer.	(a) $w^2 + 2w = 120$. Half of 2 is 1, and $1^2 = 1$. Add 1 to both sides: $w^2 + 2w + 1 = 121$, so $(w + 1)^2 = 121$. Then $w + 1 = \pm 11$, so $w = 10$ or $w = -12$. (b) $a = 1$, $b = 2$, $c = -120$. $w = \frac{-2 \pm \sqrt{(2^2 - 4(1)(-120))}}{2} = \frac{-2 \pm \sqrt{484}}{2} = \frac{-2 \pm 22}{2}$. So $w = 10$ or $w = -12$. All three methods give the same answers, $w = 10$ and $w = -12$, and the context tells us to use $w = 10$ m. (c) Factorisation is quickest when the equation has integer factor pairs, as in the hall equation. Completing the square always works and shows the geometric meaning, because we can picture a square being completed. It was useful for the Nakuru garden equation $x^2 + 6x = 48$ which does not factorise easily over integers. Here, adding 9 gives $(x + 3)^2 = 57$. The quadratic formula is a universal tool that works for any quadratic, especially when the discriminant is not an obvious perfect square. The methods are complementary tools, not competitors.
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Section 4: Interpreting solutions in another community context

In Lesson 6, a stone was thrown upward from a Mombasa cliff, with height $h(t) = -5t^2 + 20t + 25$ metres after t seconds. (a) Write the equation you must solve to find when the stone hits the sea. (b) Use the discriminant to say how many real solutions there will be, before you solve. (c) Solve the equation and state which solution you accept. Explain your choice in a Claim-Evidence-Reasoning format.	(a) The stone hits the sea when $h = 0$, so $-5t^2 + 20t + 25 = 0$. (b) $a = -5$, $b = 20$, $c = 25$. The discriminant is $b^2 - 4ac = 400 - 4(-5)(25) = 400 + 500 = 900$. Since $900 > 0$, there are two real solutions. (c) The formula gives $t = \frac{-20 \pm \sqrt{900}}{2 \times -5} = \frac{-20 \pm 30}{-10}$. So $t = -1$ or $t = 5$. Claim: The stone hits the sea after 5 seconds. Evidence: The equation gives two solutions, $t = -1$ and $t = 5$, and both make $h = 0$. Reasoning: Time cannot be negative in this situation, so $t = -1$ is rejected. This is the same as rejecting $w = -12$ in the hall problem. Only the solution that makes sense in context is accepted.
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Section 5: Answering the driving question

Write a clear explanation that answers the driving question: How can we use quadratic equations to find unknown	Quadratic equations help a community find unknown quantities when the unknown is multiplied by itself, as with area. (i) A linear equation has the variable only to the power 1 and can be solved by simple inverse operations. A quadratic equation has a squared term, as in $w^2 + 2w - 120 = 0$, and cannot be solved that way, so it is necessary for area and similar problems. (ii) For the Githurai hall, we wrote $w(w + 2) = 120$, expanded it to $w^2 + 2w - 120 = 0$, and identified $a = 1$, $b = 2$, $c = -120$. We factorised it to $(w + 12)(w$
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dimensions and quantities that help our community solve real planning problems, like building the Harambee hall in Githurai? Use the following in your answer: (i) how quadratic expressions and equations differ from linear ones, (ii) the steps of your unit journey from the hall sketch to a solution, and (iii) why interpreting solutions in context matters. Include at least one other community example from the unit.	$-10) = 0$ and used the Zero Product Property to get $w = 10$ or $w = -12$. When factorisation is hard, completing the square and the quadratic formula still work, and the discriminant tells us the number of real solutions before we solve. (iii) Both roots satisfy the equation, but a negative width is physically impossible, so the hall must be 10 m wide and 12 m long, with an area of 120 m². In the Mombasa stone problem we likewise rejected $t = -1$ and accepted $t = 5$ seconds. Other examples, such as the Nakuru school garden, Kisumu fish-drying yard or mandazi pricing, can also be modelled and solved. Knowing how to write and solve a quadratic equation lets people answer real questions about land, construction, pricing and planning, instead of guessing.
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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Modelling the situation as a quadratic equation	Writes $w(w + 2) = 120$ correctly, expands and rearranges accurately to $w^2 + 2w - 120 = 0$, identifies a, b and c, and explains clearly why the equation is not linear.	Writes and expands the equation correctly and identifies a, b and c, with a partial explanation of why it is not linear.	Has errors in forming or expanding the equation, or cannot identify a, b and c or explain how it differs from a linear equation.
Accuracy of solution methods	Solves correctly by factorisation, completing the square and the quadratic formula, with every step shown and checked, and uses the discriminant correctly.	Solves correctly with at least two methods and shows most steps, with minor slips in the other method or the discriminant.	Makes major errors in methods or steps, or uses only one method with limited accuracy.
Interpreting solutions in context	Rejects negative roots with clear reasoning, states hall dimensions of 10 m by 12 m with units, verifies the area of 120 m ² , and applies the same reasoning to the Mombasa problem.	Chooses the correct root and states the dimensions with a brief reason, with some missing checks or units.	Accepts both roots without interpretation or gives no reason for the choice.
Comparing methods and using evidence	Explains when each method is most useful with specific examples from the unit, such as the Nakuru garden, and uses Claim-Evidence-Reasoning.	Describes when each method is useful, with general or limited examples.	Gives vague or incorrect comparisons with no supporting examples.
Communicating the answer to the driving question	Gives a coherent, well-organised explanation linking the unit journey to community planning, with precise mathematical vocabulary and another	Gives a mostly clear explanation of the journey and its community use, with some gaps in vocabulary or links.	Gives a disjointed or incomplete explanation that does not link quadratics to community problems.

	Kenyan example.		
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